

Quiz 5 (Version 2)

CAS CS 132: *Geometric Algorithms*

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Name: Nathan Mull

BUID: 12345678

- ▷ You will have approximately 35 minutes to complete this exam.
- ▷ Your final solution must appear in the solution boxes for each problem. **Only include your final solution in the solution box. You must show your work outside of the solution box.** You will not receive credit if you don't show your work.

1 Determinants

$$\begin{bmatrix} 2 & 1 & 4 \\ 1 & 2 & -1 \\ 4 & 1 & 2 \end{bmatrix}$$

Determine the determinant of the above matrix using the procedure we discussed in lecture.

$$\overset{(\text{swap})}{\begin{bmatrix} 1 & 2 & -1 \\ 2 & 1 & -4 \\ 4 & 1 & 2 \end{bmatrix}} \sim \begin{bmatrix} 1 & 2 & -1 \\ 0 & -3 & 6 \\ 0 & -7 & 6 \end{bmatrix} \overset{(\text{scale})}{\sim} \begin{bmatrix} 1 & 2 & -1 \\ 0 & 1 & -2 \\ 0 & -7 & 6 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & -1 \\ 0 & 1 & -2 \\ 0 & 0 & -8 \end{bmatrix}$$

$$\det(A) = \frac{(-1)^2}{c} u_{11} u_{22} u_{33} = \frac{(-1)^1}{(-1/3)} (1)(1)(-8) = -24$$

Solution.

$$-24$$

2 Characteristic Equation

Determine a matrix in $\mathbb{R}^{2 \times 2}$ with nonzero entries whose characteristic polynomial is $\lambda^2 - 3\lambda + 10$.

$$\begin{aligned}\lambda^2 - 3\lambda + 10 &= (\lambda^2 - 3\lambda + 2) + 8 \\ &= (\lambda - 2)(\lambda - 1) + 8\end{aligned}$$

$$\begin{bmatrix} 2 & 4 \\ -2 & 1 \end{bmatrix}$$

Note: there are multiple answers

Solution.

$$\begin{bmatrix} 2 & 4 \\ -2 & 1 \end{bmatrix}$$

3 Eigenbases

$$\begin{bmatrix} 7 & -3 \\ 18 & -8 \end{bmatrix}$$
$$\begin{bmatrix} 7 & -318 & -8 \end{bmatrix}$$

Determine all eigenvalues of the above matrix, along with bases for the corresponding eigenspaces.

$$\begin{aligned} \det(A - \lambda I) &= (7 - \lambda)(-8 - \lambda) + 54 \\ &= (\lambda - 7)(\lambda + 8) + 54 \\ &= \lambda^2 + \lambda - 56 + 54 \\ &= \lambda^2 + \lambda - 2 \\ &= (\lambda + 2)(\lambda - 1) \end{aligned}$$

$$A + 2I = \begin{bmatrix} 9 & -3 \\ 18 & -6 \end{bmatrix} \sim \begin{bmatrix} 1 & -\frac{1}{3} \\ 0 & 0 \end{bmatrix} \quad \begin{array}{l} x_1 = \frac{1}{3}x_2 \\ x_2 \text{ is free} \end{array} \quad \vec{v}_1 = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$$

$$A - I = \begin{bmatrix} 6 & -3 \\ 18 & -9 \end{bmatrix} \sim \begin{bmatrix} 1 & -\frac{1}{2} \\ 0 & 0 \end{bmatrix} \quad \begin{array}{l} x_1 = \frac{1}{2}x_2 \\ x_2 \text{ is free} \end{array} \quad \vec{v}_2 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

Solution.

$$\lambda_1 = -2 \quad \text{Nul}(A + 2I) = \text{span} \left\{ \begin{bmatrix} 1 \\ 3 \end{bmatrix} \right\}$$

$$\lambda_2 = 1 \quad \text{Nul}(A - I) = \text{span} \left\{ \begin{bmatrix} 1 \\ 2 \end{bmatrix} \right\}$$